MSG No. 136.499 $P4'_2/mnm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m.m'm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,8,11,12,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,6,7,9,10,13,14]`

Table 2: Wyckoff site: `2b`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,4,5,8,11,12,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,3,6,7,9,10,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,13,14]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,15,16]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,8,9,10]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[6,7,11,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3,13,14]`
3	$[0,\ 0,\ -z]$	`[5,8,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7,9,10]`

Table 6: Wyckoff site: `4f`, site symmetry: `m.2'm'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,8,11,16]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[2,7,10,13]`
3	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,9,14]`

continued ...

Table 6

No.	position	mapping
4	$[-x, -x, 0]$	`[4,5,12,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.m'2'`

No.	position	mapping
1	$[x, -x, 0]$	`[1,8,12,15]`
2	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[2,7,9,14]`
3	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[3,6,10,13]`
4	$[-x, x, 0]$	`[4,5,11,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[5,8]`
4	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2}, 0, -z\right]$	`[11,12]`
7	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[13,14]`
8	$\left[\frac{1}{2}, 0, z\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[x, y, 0]$	`[1,8]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	`[2,7]`
3	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}\right]$	`[3,6]`
4	$[-x, -y, 0]$	`[4,5]`
5	$\left[\frac{1}{2}-y, x+\frac{1}{2}, \frac{1}{2}\right]$	`[9,14]`
6	$\left[y+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[10,13]`
7	$[y, x, 0]$	`[11,16]`
8	$[-y, -x, 0]$	`[12,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m'`

No.	position	mapping
1	$[x, x, z]$	`[1,16]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[2,13]`
3	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[3,14]`
4	$[-x, -x, z]$	`[4,15]`
5	$[-x, -x, -z]$	`[5,12]`

continued ...

Table 10

No.	position	mapping
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[6,9]`
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[7,10]`
8	$[x,\, x,\, -z]$	`[8,11]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[-x,\, -y,\, -z]$	`[5]`
6	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[7]`
8	$[x,\, y,\, -z]$	`[8]`
9	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[10]`
11	$[y,\, x,\, -z]$	`[11]`
12	$[-y,\, -x,\, -z]$	`[12]`
13	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[14]`
15	$[-y,\, -x,\, z]$	`[15]`
16	$[y,\, x,\, z]$	`[16]`